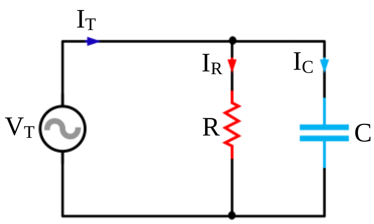
AC Circuit Part 3

Parallel AC Circuits

continuation....

When dealing with series AC circuits, use the current phasor as the reference phasor since the same current flows through all the components. The phasor sum of the various voltage drops is equal to the applied voltage.

Parallel RC Circuit



Total current, I_T

$$I_T = I_R + jI_C = |I_T|\angle\theta$$
$$|I_T| = \sqrt{{I_R}^2 + {I_C}^2}$$
$$\theta = \tan^{-1}\frac{I_C}{I_R}$$

Total admittance, Y

$$Y = G + jB_C = |Y|\angle\theta$$
$$|Y| = \sqrt{G^2 + {B_C}^2}$$
$$\theta = \tan^{-1}\frac{B_C}{G}$$

Total impedance, Z $\quad Z = \frac{1}{Y}\theta$

Total voltage, V_T $\quad V_T = I_T Z$

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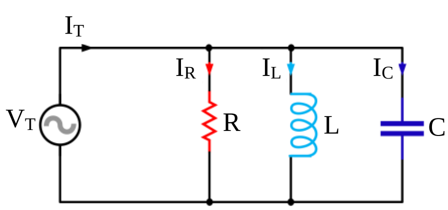
Parallel RLC Circuit



Total current, I_T

$$I_T = I_R + jI_C - jI_L = |I_T|\angle\theta$$
$$|I_T| = \sqrt{{I_R}^2 + (I_C - I_L)^2}$$
$$\theta = \pm\tan^{-1}\frac{(I_C - I_L)}{I_R}$$

continue

Total admittance, Y

$$Y = G + jB_C - jB_L = |Y|\angle\theta$$
$$|Y| = \sqrt{G^2 + (B_C - B_L)^2}$$
$$\theta = \pm\tan^{-1}\frac{(B_C - B_L)}{G}$$

Total impedance, Z $\quad Z = \frac{1}{Y}$

Total voltage, V_T $\quad V_T = I_T Z$

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Powers of AC Circuits

True/Real/Average/ Active Power
It is the component of apparent power that represents true work. It is expressed in **watts (W)** and is equal to voltamperes multiplied by the power factor.

$$P = {I_R}^2 R \qquad P = \frac{{V_R}^2}{R}$$
$$P = I_R V_R \qquad P = V_T I_T \cos\theta$$

Note: cos θ = Power Factor (PF)

Reactive Power
It is the power value obtained by multiplying together the effective value of current in amperes, the effective value of voltage in volts and the sine of the angular phase difference between current and voltage. Also called reactive voltamperes and wattless power. It is expressed in **volt-ampere-reactive (VAR).**

$$Q = I_X X_{eq} = \frac{{V_X}^2}{X_{eq}} = I_X V_X = V_T I_T \sin\theta$$

Notes:
I_x is net reactive current (I_c - I_L).
V_x is net reactive voltage (V_L - V_C).
X_{eq} is equivalent reactance (X_L - X_C).
Sin θ = Reactive Factor

Apparent Power
It is the power value obtained by multiplying the effective values of voltage and current. It is expressed in **voltamperes (VA).**

$$S = {I_T}^2 Z = \frac{{V_T}^2}{Z} = V_T I_T$$

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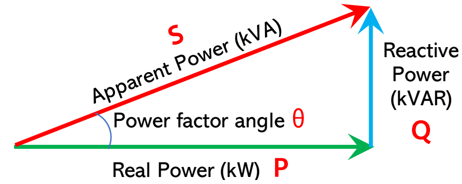
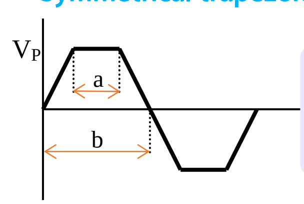
POWER TRIANGLE



$$\cos\theta = \frac{P}{S} = \text{Power Factor (PF)}$$
$$\sin\theta = \frac{Q}{S} = \text{Reactive Factor (RF)}$$
$$S = P \pm jQ = |S|\angle\theta$$
$$|S| = \sqrt{P^2 + Q^2} \qquad \theta = \pm\tan^{-1}\frac{Q}{P}$$

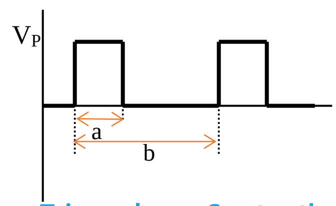
Other Alternating Waveforms

Symmetrical trapezoid



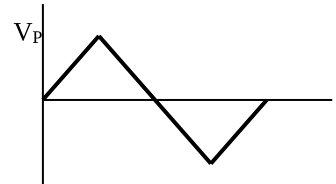
$$V_{rms} = \frac{a + 0.577(b-a)}{b}V_P$$
$$|V_{avg}| = \frac{a+b}{2b}V_P$$

DC Pulse



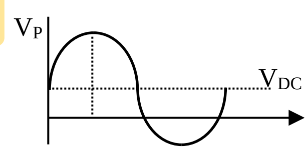
$$|V_{rms}| = V_P\sqrt{\frac{a}{b}}$$
$$|V_{avg}| = V_P\frac{a}{b}$$

Triangular or Sawtooth



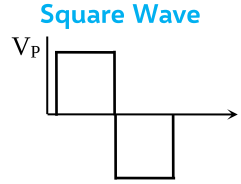
$$V_{rms} = 0.577V_P$$
$$|V_{avg}| = 0.500V_P$$

Sine Wave on DC Level



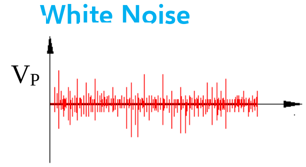
$$V_{rms} = \sqrt{{V_{DC}}^2 + \frac{{V_P}^2}{2}}$$

Square Wave



$$V_{rms} = V_P$$
$$|V_{avg}| = V_P$$

White Noise



$$V_{rms} \approx \frac{1}{4}V_P$$

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